



An experiment a day

You can attempt to do one DIY experiment every day for a year. This book contains 365 experiments! <https://digit.in/mar20-13>



Strange cohabitation

A researcher was so interested in sand flea development, she let a bug live inside her foot for 2 months. <https://digit.in/mar20-14>

Beverly Clock

Running since 1864

University of Otago, New Zealand

This is a clock designed by New Zealand watchmaker and astronomer, Arthur Beverly. Since the time of its construction, it has been ticking. It is an atmospheric clock that uses the daily variations in the temperature and pressure as its energy source. There have been periods where the clock has stopped working when the temperature and pressure differences were not sufficient. However, when the differences became more pronounced, the clock resumed ticking. The Beverly clock is sometimes referred to as a perpetual motion machine, although technically this is inaccurate. The Swiss watchmaker Jaeger-LeCoultre offers a similar mechanism in its Atmos range of clocks, with prices starting at around ₹5 lakh.

Credit: amanderson2



Beal germination experiment

Running since 1879

Michigan State University, US

William J Beal started an experiment to answer a question that local farmers had asked him, essentially for how long can seeds remain dormant in soil. So, Beal filled jars with soil and seeds, sealed them, and buried them in the ground. Every five years, he would remove a jar and germinate the seeds, to check if they grew. Beal opened up six jars before he retired, and the work was carried on by his colleague. The caretakers of the experiment started increasing the gaps between opening the bottles, first to a decade then to two. The last bottle was opened in 2000, and only two seeds managed to grow. The next bottle is scheduled to be opened this year. The experiment is scheduled for completion in 2100.



Old Rotation

Running since 1896

Auburn University, US

The old rotation is one of the longest running field experiments in the world, and was the first experiment to prove that alternating cotton crops with legumes could introduce more nitrogen into the soil, which would be beneficial for the cotton crops. In alternate seasons, cotton and legumes are planted, to demonstrate the long term viability of using such a system. There are 13 plots each with a different system. These are cotton plants with and without legume rotation, an additional corn rotation as well as with and without fertiliser. One plot rotates cotton with corn, legumes and soybeans. The experiment has been running continuously for over 124 years, and clearly shows that the rotated cotton crops yield more.

The Pitch Drop Experiment

Running since 1927

University of Queensland, Australia

Professor Thomas Parnell wanted to demonstrate that some substances that appeared solid were actually very viscous liquids, and devised a sealed experiment that uses a funnel to slowly drop a glob of pitch into a beaker. Once about every ten years, a drop of the pitch breaks off and falls into the beaker. There have been nine drops so far, and everyone who is invested in the experiment are eagerly waiting for the tenth drop. There is enough pitch in the funnel to last for another century or so. The pitch drop experiment is considered to be the longest running laboratory experiment in the world, because the Beverly Clock has been known to stop on occasion, and the Oxford Bell was interrupted while servicing the apparatus.

